

Estimation of pure component properties

Part 1. Estimation of the normal boiling point of non-electrolyte organic compounds via group contributions and group interactions

Yash Nannoolal^a, Jürgen Rarey^{a,b,*}, Deresh Ramjugernath^a, Wilfried Cordes^b

^a School of Chemical Engineering, University of Kwa-Zulu Natal, Durban 4041, South Africa

^b DDBST GmbH, Industriestr. 1, 26121 Oldenburg, FRG

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Abstract

An improved group contribution method for the estimation of the normal boiling point of non-electrolyte organic compounds was developed using experimental data for approximately 2850 components stored in the Dortmund Data Bank (DDB). The new model is based on a method published earlier, which was extended using a steric parameter, additional corrections and group interaction parameters as well as several additional groups for extended range of applicability. The mathematical formalism was modified to allow for separate examination and regression of individual contributions. The results of this separate examination lead to the detection of unreliable data and the definition of further structural groups. For the boiling point prediction, only the molecular structure of the compound is used. The results of the new method are compared to the previous work and several currently used methods and are shown to be far more accurate and reliable. The detailed comparison for a large number of different sets of chemically similar compounds allows one to access the probable estimation error of the different methods. Structural groups were defined in a standardized form and fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions.

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1. Introduction

The basis for any design and simulation of chemical, biochemical and environmental systems (e.g. a chemical plant) is a reliable set of physical and chemical pure component and mixture properties. As it is not always possible to find experimental values in the literature and since measurement is expensive and time consuming or sometimes even difficult or impossible, estimation methods are generally of great value.

In principle, properties of a pure component can be derived from the structure of the molecule, and, in some cases, state variables, such as temperature and pressure. For the property

estimation of pure components, group contribution methods have been widely used. The properties of a compound are calculated as a function of structurally dependent parameters, which can be determined by summing the number frequency of each group multiplied by its contribution, based on the assumption that the effects of the individual groups are additive. These methods have the advantage of supplying quick estimates with only the structure of the component known, but are often of questionable and unknown reliability.

This paper presents a significant improvement and extension of applicability of a method for the estimation of normal boiling points by group contribution published earlier [1].

Most authors start with methods for estimation of the critical point. At the critical point, changes of the attrac-

* Corresponding author. Tel.: +49 441 798 3846; fax: +49 441 798 3330.
E-mail address: rarey@ddbst.de (J. Rarey).

tive and repulsive part of the equation of state compensate when changing the volume ($(\partial P/\partial V)_{T_c, P_c} = 0$). This behavior has a direct relation to the intermolecular potential. On the other hand, in the near critical region both liquid and vapor, which become identical at the critical point, behave like dense gases with the chemical potential in both phases far from any simple reference state. Calculation of the properties of far sub-critical liquid phases usually present in technical systems is very insecure. In addition, only critical data for a few hundred rather simple and stable molecules are available.

The normal boiling point was chosen as the first and most important property, as it gives information about the chemical potential of a molecule in a rather cool and structured liquid and because a large amount of data are available for molecules which are very different in size and chemical structure. The boiling point is governed by the “equilibrium” between enthalpic and entropic effects ($\Delta H_v(T_b) = T_b \Delta S_v(T_b)$). At the normal boiling point, ΔH_v and ΔS_v are usually close to those of the ideal gas, which serves as a convenient reference state. ΔS_v is predominantly due to the gain in translational freedom when going from the liquid to the (approximately ideal) vapor phase^{1,2}. ΔH_v results from the amount (proportional to the molecular surface) and nature (dispersion, polar–non-polar, polar–polar, etc.) of the intermolecular contacts.

2. Available methods for the prediction of normal boiling points

A variety of estimation methods for normal boiling points from molecular structure are available. A broad overview on these methods was given by Poling et al. [3] and a brief review of melting and boiling point methods by Boethling and Mackay [4]. Several methods are restricted to individual classes of substances, while others require the knowledge of molecular descriptors usually obtained from quantum mechanical methods. As in the previous work, we will compare our results with those of Constantino and Gani (CG) [5], Joback and Reid (JR) [6], Stein and Brown (SB) [7] and Marrero-Morejon and Pardillo-Fontdevila (MP) [8]. All of these methods have in common the fact that they only require the knowledge about the molecular structure and therefore are comparable. As the original paper of Marrero-Morejon and Pardillo-Fontdevila [8] contains several misprints, we have used in this work the corrected parameters supplied by the

authors and published by Poling et al. [3]. A critical discussion of these methods can be found in the previous work [1].

Two other methods recently published are that of Marrero-Morejon and Gani (MG1) [9] and Ericksen and coworkers (EWOR) [10]. In MG1, Marrero-Morejon and Gani employ a multi-level approach based, in the case of T_b -estimation on 167 first-order, 106 second-order and 51 third-order (in total 324) contributions. The authors report an absolute average deviation of 5.89 K for a set of 1794 components using approximately 5.5 data points per adjustable parameter, but report no error distribution or a test of the predictive capability using data not included in the regression. As in the CG-method, they employ a logarithmic dependence of T_b on the sum of group contributions, which leads to large errors for small and large molecules, e.g. in case of *n*-alkanes, using the published parameters. We therefore did not include this later version in the comparison.

The EWOR-method (EW) uses a similar large number of group contributions (>300) on a significantly smaller set of data. An absolute average deviation of 13 K is reported for a test set of 384 components not included in the regression. EWOR employs an equation that includes also the van der Waals volume of the molecule as a parameter. We decided to include this method in the comparison for all components for which we were able to estimate the van der Waals volume.

3. Development of the new method and results

From the experience gained with the development of the previous method [1], several requirements for improvement became apparent and are discussed here. The list of structural groups for the new method and the list of second-order corrections are given in Tables 1 and 2, and the group and correction parameters are found in Tables 3 and 4. For convenience, the group and correction parameters are also included in Tables 1 and 2. Table 5 lists the group interaction contributions discussed below. The model Eq. (1) is used for the calculation of T_b . In case of the group interactions for multifunctional components, the interaction frequency is explained in the respective section.

3.1. Regression algorithm

In the previous method, a linear regression of the equation

$$f(T_b) = a_0 + \sum N_i C_i \quad \text{and} \quad T_B = \frac{\sum N_i C_i}{n^a + b} + c \quad (1)$$

with, N_i the number of groups of type *i*; C_i the group contribution of group *i* [K]; a_0 the constant value from linear regression; *a*, *b*, *c* the adjustable parameters ($a = 0.6583$, $b = 1.6868$, $c = 84.3395$); and *n* the number of atoms in the molecule (except hydrogen), was employed. The linearization function

¹ “Some gas-phase molecules, like CH₄, have, for example, a negligible entropy due to rotation and vibration. Others, like CCl₄, have a large contribution from this motion; about half the entropy of gaseous CCl₄ is due to rotational and vibrational contributions. But molecules like CH₄ and CCl₄ seem to obey Trouton’s rule equally well. Thus, it seems that no major changes in the rotational and vibrational entropies are occurring.” Barrow [2], p. 232.

² As discussed for hexane and cyclohexane by Cordes and Rarey [1].

Table 1

Group definitions (ID: identification number, PR: priority)

Description		Name, contribution [K]	ID	PR	Occurs, e.g. in
Periodic group 17					
Fluorine					
F—	F— connected to C or Si	F—(C, Si), 106.5492	19	86	2-Fluoropropane, trimethylfluorosilane
	F— connected to a C or Si already substituted with one F or Cl and one other atom	F—(C—([F, Cl]))—a, 78.7578	22	83	1-Chloro-1,2,2,2-tetrafluoroethane[R124], difluoromethylsilane
	F— connected to C or Si already substituted with at least one F and two other atoms	F—(C—([F, Cl]))—b, 53.1871	21	80	1,1,1-Trifluoroethane, 2,2,3,3-tetrafluoropropionic acid
	F— connected to C or Si already substituted with at least one Cl and two other atoms	F—(C—(Cl))—b, 111.0590	102	81	Trichlorofluoromethane[R11], 1,1-dichloro-1-fluoroethane [R141B]
	F— connected to C or Si already substituted with two F or Cl	F—(C—([F, Cl]2)), 103.5672	23	82	1',1',1'-Trifluorotoluene, 2,2,2-trifluoroethanol, trifluoroacetic acid
	F— connected to an aromatic carbon	F—(C(a))—, 19.5575	24	85	Fluorobenzene, 4-fluoroaniline
	F— on a C=C (vinylfluoride)	—CF=C<, 49.2701	20	84	Vinyl fluoride, trifluoroethene, perfluoropropylene
Chlorine					
Cl—	Cl— connected to C or Si not already substituted with F or Cl	Cl—(C, Si), 330.9117	25	72	Butyl chloride, 2-chloroethanol, chloroacetic acid
	Cl— connected to C or Si already substituted with one F or Cl	Cl—((C, Si)—([F, Cl])), 287.1863	26	71	Dichloromethane, dichloroacetic acid, dichlorosilane
	Cl— connected to C or Si already substituted with at least two F or Cl	Cl—((C, Si)—([F, Cl]2)), 267.4170	27	69	Ethyl trichloroacetate, trichloroacetonitrile
	Cl— connected to an aromatic C	Cl—(C(a)), 205.7363	28	73	Chlorobenzene
	Cl— on a C=C (vinylchloride)	—CCl=C<, 292.5816	29	70	Vinyl chloride
COCl—	COCl— connected to C (acid chloride)	COCl—, 778.9151	77	19	Acetyl chloride, phenylacetic acid chloride
Bromine					
Br—	Br— connected to a non-aromatic C or Si	Br—(C/Si(na)), 419.4959	30	66	Ethyl bromide, bromoacetone
Br—	Br— connected to an aromatic C	Br—(C(a)), 377.6775	31	67	Bromobenzene
Iodine					
I—	I— connected to C or Si	I—(C, Si), 556.3944	32	64	Ethyl iodide 2-iodotoluene
Periodic group 16					
Oxygen					
—OH	—OH for aliphatic chains with less than five C (cannot be connected to aromatic fragments)	—OH short chain< C ₅ , 488.0819	36	91	Ethanol, propanediol
	—OH connected to C or Si substituted with one C or Si in an at least five C or Si containing chain (primary alcohols)	—OH> C ₄ , 443.8712	35	87	1-Nonanol, tetrahydrofurfuryl alcohol, ethylene cyanohydrin
	—OH connected to a C or Si substituted with two C or Si in a at least three C or Si containing chain (secondary alcohols)	HO—((C, Si)2H—(C, Si)—(C, Si)—), 390.2446	34	89	2-Butanol, cycloheptanol
	—OH connected to C which has four non-hydrogen neighbors (tertiary alcohols)	—OH tert, 349.9409	33	90	Tert-butanol, diacetone alcohol
	—OH connected to an aromatic C (phenols)	—OH (Ca), 361.4775	37	88	Phenol, methyl salicylate
—O—	—O— connected to two neighbors which are each either C or Si (ethers)	(C, Si)—O—(C, Si), 146.4836	38	93	Diethyl ether, 1,4-dioxane
	—O— in an aromatic ring with aromatic C as neighbors	(C(a))—O(a)—(C(a)), 37.1936	65	92	Furan, furfural
—CHO	CHO— connected to non-aromatic C (aldehydes)	CHO—(Can), 553.8090	52	53	Acetaldehyde, pentanedial
	CHO— connected to aromatic C (aldehydes)	CHO—(Ca), 560.1024	90	52	Furfural, benzaldehyde

Table 1 (Continued)

	Description	Name, contribution [K]	ID	PR	Occurs, e.g. in
> C=O	—CO— connected to two non-aromatic C (ketones)	O=C< (Cna)2, 618.9782	51	55	Acetone, methyl cyclopropyl ketone
	—CO— connected to two C with atleast one aromatic C (ketones)	(O=C< (C)2)a, 606.1797	92	54	Acetophenone, benzophenone
	—CO connected to N	> N(C=O), 492.0707	109	39	Methyl thioacetate
	—CO connected to two N (urea)	> N—(C=O)—N<, 1045.0343	100	2	Urea-1,1,3,3-tetramethyl
O=C(—O—)2	Non-cyclic carbonate	O=C(—O—) 2, 879.7062	79	15	Dimethyl carbonate
COOH—	—COOH connected to C	COOH —(C), 1080.3139	44	24	Acetic acid
—COO—	HCOO— connected to C (formic acid ester)	HCOO —(C), 642.0427	46	27	Ethyl formate, phenyl formate
	—COO— connected to two C (ester)	(C)—COO —(C), 636.2020	45	25	Ethyl acetate, vinyl acetate
	—COO— in a ring, C is connected to C (lactone)	—C(c)OO—, 1142.6119	47	26	ε-Caprolactone, crotonolactone
—OCOO—	—CO connected to two O (carbonates)	—OCOO—, 1573.3769	103	34	Propylene carbonate 1,3-dioxolan-2-one
—OCON<	—CO connected to O and N (carbamate)	—OCON<, 886.7613	99	1	Trimethylsilyl methylcarbamate
> (OC2)<	> (OC2)< (epoxide)	> (OC2)<, 820.7118	39	50	Propylene oxide
—CO—O—CO—	Anhydride connected to two C	—C=O—O—C=O—, 1251.2675	76	12	Acetic anhydride, butyric anhydride
	Cyclic anhydride connected to two C	(—C=O—O—C=O—)r, 2082.3288	96	11	Maleic anhydride, phthalic anhydride
—O—O—	Peroxide	—O—O—, 273.1755	94	32	Di-tert-Butylperoxide
Sulfur					
—S—S—	—S—S— (disulfide) connected to two C	(C)—S—S—(C), 864.5074	55	51	Dimethyldisulfide, 1,2-dicyclopentyl-1,2-disulfide
—SH	—SH connected to C (thioles)	SH—(C), 434.0811	53	74	1-Propanethiol
—S—	—S— connected to two C	(C)—S—(C), 461.5784	54	75	Methyl ethyl sulfide
	—S— in an aromatic ring	—S(a)—, 304.3321	56	76	Thiazole, thiophene
—SO2—	Non-cyclic sulfone connected to two C (sulfones)	(C)—SO2—(C), 1559.9840	82	18	Sulfolane, divinylsulfone
> SO4	S(=O)2 connected to two O (sulfates)	> SO4, 1483.1289	104	35	Dimethyl sulfate
—SO2N<	—S(=O)2 connected to N	—SO2N<, 1506.8136	105	36	N,N-diethylmethanesulfonamide
> S=O	Sulfoxide	> S=O, 1379.4485	107	37	1,4-Thioxane-S-oxide tetramethylene sulfoxide
SCN—	SCN— (thiocyanate) connected to C	SCN—(C), 1018.4865	81	20	Allyl isothiocyanate
Selenium					
> Se<	> Se< connected to atleast one C or Si	> Se<, 562.1791	116	46	Dimethyl selenide
Periodic group 15					
Nitrogen					
NH2—	NH2— connected to either C or Si	NH2—(C, Si), 321.1759	40	95	Hexylamine, ethylenediamine
	NH2— connected to an aromatic C	NH2— (Ca), 441.4388	41	94	Aniline, benzidine
—NH—	—NH— connected to two neighbors which are each either C or Si (secondary amines)	(C, Si)—NH—(C, Si), 223.0992	42	99	Diethylamine, diallyl amine
	—NH— connected to two C or Si neighbors, with atleast one aromatic neighbor (secondary amines)	(C, Si)a—NH—(Ca, Si)a, 201.3224	97	98	Morpholine pyrrolidine
> N<	> N— connected to three neighbors which are each either C or Si (tertiary amines)	(C, Si)2> N—(C, Si), 126.2952	43	100	N,N-dimethylaniline, nicotine
	Quaternary amine connected to four C or Si	(C, Si)2> N< (C, Si)2, —109.6269	101	33	N,N,N,N-tetramethylmethylenediamine
=N—	Double bonded amine connected to atleast one C or Si	(C, Si)=N—, 229.2288	91	101	Acetonin
=N—	Aromatic —N— in a five-membered ring, free electron pair	=N(a)— (r5), 453.3397	66	97	Piperidine, thiazole
=N—	Aromatic =N— in a six-membered ring	=N(a)— (r6), 306.7139	67	96	Pyridine, nicotine

Table 1 (Continued)

	Description	Name, contribution [K]	ID	PR	Occurs, e.g. in
C≡N—	—C≡N (cyanide) connected to C	(C)—C≡N, 719.2462	57	56	Acetonitrile, 2,2'-dicyano diethyl sulfide
	—C≡N (cyanide) connected to N	(N)—C≡N, 971.0365	111	41	Dimethylcyanamide
	—C≡N (cyanide) connected to S	(S)—C≡N, 659.7336	108	38	Methyl thiocyanate
CNCNC—r	Imadizole	...=CNC=NC=... , 484.6371	106	3	1-Methyl-1-imadizole
—CONH<	—CONH2 (amide)	—CONH2, 1487.4109	50	28	Acetamide
	—CONH— (monosubstituted amide)	—CONH—, 1364.5333	49	48	N-methylformamide, 6-caprolactam
	—CON< (disubstituted amide)	—CON<, 1052.6072	48	49	N,N-dimethylformamide (DMF)
OCN—	OCN— connected to C or Si (cyanate)	OCN—, 660.4645	80	29	Butylisocyanate, hexamethylene diisocyanate
ONC—	ONC— (oxime)	ONC—, 1041.0851	75	30	Methyl ethyl ketoxime
—ON=	—ON= connected to C or Si (isoazole)	—ON=(C, Si), 612.9506	115	45	Isoazole 5-phenyl isoazole
NO2—	Nitrites (esters of nitrous acid)	O=N—O—(C), 494.2668	74	23	Ethyl nitrite, nitrous acid methyl ester
	NO2— connected to aliphatic C	NO2—(C), 866.5843	68	21	1-Nitropropane
	NO2— connected to aromatic C	NO2—(C(a)), 821.4141	69	22	Nitrobenzene
NO3—	Nitrate (esters of nitric acid)	NO3—, 920.3617	72	14	N-butyl nitrate, 1,2-propanediol dinitrate
Phosphorous					
>P(O—)3	Phosphates with four O substituents	PO(O—)3, 1153.1344	73	10	Triethyl phosphate, tris-(2,4-dimethylphenyl) phosphate
>P<	Phosphorus connected to atleast 1 C or S (phosphine)	>P<, 428.8911	113	43	Triphenylphosphine triethylphosphane
Arsine					
AsCl2—	AsCl2 connected to C	AsCl2—, 1149.9670	84	17	Ethylarsenic dichloride
Periodic group 14					
Carbon					
—CH3	CH3— not connected to either N, O, F or Cl	CH3—(ne), 177.3066	1	104	Decane
	CH3— connected to either N, O, F or Cl	CH3—(e), 251.8338	2	102	Dimethoxymethane, methyl butyl ether
	CH3— connected to an aromatic atom (not necessarily C)	CH3—(a), 157.9527	3	103	Toluene, <i>p</i> -methyl-styrene
—CH2—	—CH2— in a chain	—C(c)H2—, 239.4531	4	111	Butane
	—CH2— in a ring	—C(r)H2—, 239.4957	9	112	Cyclopentane
>CH—	>CH— in a chain	>C(c)H—, 240.6785	5	117	2-Methylpentane
	>CH— in a ring	>C(r)H—, 222.1163	10	116	Methylcyclohexane
>C<	>C< in a chain	>C(c)<, 249.5809	6	119	Neopentane
	>C< in a chain connected to at least one aromatic carbon	>C(c)<(a), 201.0115	8	108	Ethylbenzene, diphenylmethane
	>C< in a chain connected to at least one F, Cl, N or O	>C(c)<(e), 266.8769	7	107	Ethanol
	>C< in a ring	>C(r)<, 209.9749	11	118	beta-pinene
	>C< in a ring connected to at least one aromatic carbon	>C(r)<(Ca), 244.3581	14	106	Indene, 2-methyl tetralin
	>C< in a ring connected to at least one N or O which are not part of the ring or one Cl or F	>C(r)<(e, c), 492.0707	12	109	Cyclopentanol, menthol
	>C< in a ring connected to at least one N or O which are part of the ring	>C(r)<(e, r), 291.2291	13	110	Morpholine, nicotine
	Aromatic =CH—	=C(a)H—, 235.3462	15	105	Benzene
=C(a)<	Aromatic =C< not connected to either O, N, Cl or F	=C(a)<(ne), 315.4128	16	115	Ethylbenzene, benzaldehyde
	Aromatic =C< with three aromatic neighbors	(a)=C(a)< 2(a), 367.9649	18	114	Naphthalene, quinoline
	Aromatic =C< connected to either O, N, Cl or F	=C(a)<(e), 348.2779	17	113	Aniline, phenol
	H2C=C< (1-ene)	H2C(c)=C<, 412.6276	61	58	1-Hexene

Table 1 (Continued)

	Description	Name, contribution [K]	ID	PR	Occurs, e.g. in
	$>C=C<$ (both C have at least one non-H neighbor)	$>C(c)=C(c)<$, 475.7958	58	63	2-Heptene, mesityl oxide
	Non-cyclic $>C=C<$ connected to at least one aromatic C	$>C(c)=C(c)<$ (C(a)), 586.1413	59	60	Isosafrole, cinnamic alcohol
	Cyclic $>C=C<$	$>C(r)=C(r)<$, 475.9623	62	61	Cyclopentadiene
	Non-cyclic $>C=C<$ substituted with at least one F, Cl, N or O	$-(e)C(c)=C(c)<$, 500.2434	60	59	<i>trans</i> -1,2-dichloroethylene, perfluoroisoprene
$-C\equiv C-$	$HC\equiv C-$ (1-ine)	$HC\equiv C-$, 422.2307	64	57	1-Heptyne
	$-C\equiv C-$	$-C\equiv C-$, 512.2893	63	62	2-Octyne
$>C=C=C<$	Cumulated double bond	$>C=C=C<$, 664.0903	87	6	1,2-Butadiene dimethyl allene
$>C=C-C=C<$	Conjugated double bond in a ring	$>C=C-C=C<$, 957.6388	88	7	Cyclopentadiene abietic acid
$>C=C-C=C<$	Conjugated double bond in a chain	$>C=C-C=C<$, 928.9954	89	8	Isoprene 1,3-hexadiene
$-C\equiv C-C\equiv C-$	Conjugated triple bond	$-C\equiv C-C\equiv C-$, 1218.1878	95	9	2,4-Hexadiyne
Silicon					
$>Si<$	$>Si<$	$>Si<$, 282.0181	70	79	Butylsilane
	$>Si<$ connected to at least one O	$>Si<$ (O), 207.9312	71	77	Hexamethyl disiloxane
	$>Si<$ connected to at least one F or Cl	$>Si<$ (F, Cl), 540.0895	78	16	Trichlorosilane
Germanium					
$>Ge<$	$>Ge<$ connected to four carbons	(C) $2>Ge<$ (C) 2 , 347.7717	86	68	Tetramethylgermane
$GeCl_3-$	$GeCl_3-$ connected to carbons	$GeCl_3-$, 1209.2972	85	13	Fluorodimethylsilyl(trichloro-germanyl)methane
Stannium					
$>Sn<$	$>Sn<$ connected to four carbons	(C) $2>Sn<$ (C) 2 , 510.4223	83	65	Tetramethylstannane
Periodic group 13					
Bor					
$B(O-)_3$	Non-cyclic boric acid ester	$B(O-)_3$, 540.0895	78	16	Triethyl borate
Aluminum					
$>Al<$	$>Al<$ connected to atleast one C or Si	$>Al<$, 761.6006	117	47	Triethylaluminum

Abbreviations: (e): very electronegative neighbors (N, O, F, Cl); (ne): not very electronegative neighbors (not N, O, F, Cl); (na): non-aromatic atom or neighbor; (a): aromatic atom or neighbor; (c): atom or neighbor is part of a chain; (r): atom or neighbor is part of a ring.

Table 2

Second-order groups and corrections for normal boiling points

Name contribution [K]	Description	ID	Occurs, e.g. in
$C=C-C=O$, 40.4205	$-C=O$ connected to sp_2 carbon	118	Benzaldehyde furfural
$(C=O)-C([F, Cl]_{2,3})$, -82.2328	Carbonyl connected to carbon with two or more halogens	119	Dichloroacetyl chloride
$(C=O)-(C([F, Cl]_{2,3}))_2$, -247.8893	Carbonyl connected to two carbon with two or more halogens each	120	Perfluoro-2-propanone
$C-[F, Cl]_3$, -20.3996	Carbon with three halogens	121	1,1,1-Trifluorotoluene
$(C)_2-C-[F, Cl]_2$, 15.4720	Secondary carbon with two halogens	122	2,2-Dichloropropane
No hydrogen, -172.4201	Component has no hydrogen	123	Perfluoro compounds
One hydrogen, -99.8035	Component has one hydrogen	124	Nonfluorobutane
3/4 Ring, -62.3740	A three- or four-membered non-aromatic ring	125	Cyclobutene
Five-ring, -40.0058	A five-membered non-aromatic ring	126	Cyclopentane
<i>Ortho</i> pair(s), -27.2705	<i>Ortho</i> -position—counted only once and only if there are no <i>meta</i> or <i>para</i> pairs	127	<i>o</i> -Xylene
<i>Meta</i> pair(s), -3.5075	<i>Meta</i> -position—counted only once and only if there are no <i>para</i> or <i>ortho</i> pairs	128	<i>m</i> -Xylene
<i>Para</i> pair(s), 16.1061	<i>Para</i> -position—counted only once and only if there are no <i>meta</i> or <i>ortho</i> pairs	129	<i>p</i> -Xylene
$((C=)(C)C-CC_3)$, 25.8348	Carbon with four carbon neighbors and one double bonded carbon neighbor	130	<i>Tert</i> -butylbenzene
C_2C-CC_2 , 35.8330	Carbon with four carbon neighbors, two on each side	131	Bicyclohexyl
C_3C-CC_2 , 51.9098	Carbon with five carbon neighbors	132	Ethyl bornyl ether
C_3C-CC_3 , 111.8372	Carbon with six carbon neighbors	133	2,2,3,3-Tetramethylbutane

Table 3

Group contributions, number of components used for regressing these values and mean absolute deviation in temperature for these components

Group number	Group contribution (K)	Mean absolute error (%)	Mean absolute error (K)	Standard deviation (K)	Number of components
1	177.3066	1.36	5.86	7.89	1841
2	251.8338	1.63	6.97	9.36	267
3	157.9527	1.24	5.89	7.74	168
4	239.4531	1.20	5.40	7.12	1153
5	240.6785	1.15	5.29	7.04	391
6	249.5809	1.79	7.63	10.76	98
7	266.8769	1.66	7.01	9.30	1024
8	201.0115	1.06	5.55	7.03	183
9	239.4957	1.59	6.68	8.92	330
10	222.1163	1.28	5.51	7.50	156
11	209.9749	0.99	4.30	6.47	50
12	250.9584	1.36	5.63	8.04	78
13	291.2291	1.91	8.25	10.92	124
14	244.3581	1.38	7.25	9.89	25
15	235.3462	1.34	6.76	8.93	684
16	315.4128	1.26	6.43	8.57	539
17	348.2779	1.52	7.44	9.78	289
18	367.9649	1.09	6.11	7.82	64
19	106.5492	1.90	6.48	9.62	39
20	49.2701	3.80	8.84	10.07	9
21	53.1871	2.46	8.43	10.70	150
22	78.7578	2.48	8.26	10.54	29
23	103.5672	1.75	4.43	5.90	3
24	−19.5575	2.02	7.88	9.93	28
25	330.9117	2.00	8.23	10.71	120
26	287.1863	1.76	6.90	9.23	49
27	267.4170	2.21	8.59	11.53	55
28	205.7363	1.81	8.78	11.39	73
29	292.5816	1.98	6.90	9.29	35
30	419.4959	2.14	8.06	9.95	68
31	377.6775	1.29	6.24	8.40	26
32	556.3944	1.42	5.81	6.96	28
33	349.9409	1.45	6.30	8.49	49
34	390.2446	1.74	7.99	10.25	97
35	443.8712	1.44	7.16	9.35	89
36	488.0819	2.12	9.20	11.61	53
37	361.4775	2.07	10.38	12.98	63
38	146.4836	1.69	7.36	9.81	458
39	820.7118	1.15	4.63	6.49	20
40	321.1759	1.16	4.88	6.11	55
41	441.4388	1.21	6.31	7.77	38
42	223.0992	0.87	3.66	5.21	41
43	126.2952	2.13	9.49	12.13	51
44	1080.3139	1.67	8.56	11.29	60
45	636.2020	1.64	7.76	10.37	284
46	642.0427	1.06	4.52	8.22	19
47	1142.6119	2.55	12.32	14.67	4
48	1052.6072	1.87	8.86	11.88	10
49	1364.5333	1.50	7.53	10.20	6
50	1487.4109	1.50	7.40	8.14	4
51	618.9782	1.69	7.63	10.22	114
52	553.8090	1.79	7.24	10.44	36
53	434.0811	0.98	3.79	4.81	50
54	461.5784	1.35	5.56	7.10	56
55	864.5074	0.71	2.96	3.90	4
56	304.3321	1.47	6.84	9.77	31
57	719.2462	1.26	5.71	7.79	44
58	475.7958	1.53	6.11	8.04	115
59	586.1413	1.49	7.82	10.01	12
60	500.2434	2.05	7.10	8.87	29
61	412.6276	1.60	5.93	7.73	198

Table 3 (Continued)

Group number	Group contribution (K)	Mean absolute error (%)	Mean absolute error (K)	Standard deviation (K)	Number of components
62	475.9623	2.04	8.44	10.71	91
63	512.2893	0.67	2.74	3.68	24
64	422.2307	1.80	6.23	8.11	28
65	37.1936	0.64	2.65	3.75	18
66	453.3397	1.52	6.82	8.27	23
67	306.7139	1.49	7.03	8.72	44
68	866.5843	4.25	16.36	18.45	5
69	821.4141	1.15	6.12	7.90	30
70	282.0181	1.40	5.01	7.11	37
71	207.9312	2.25	10.76	14.25	43
72	920.3617	0.86	3.49	4.03	6
73	1153.1344	0.90	5.05	5.18	4
74	494.2668	0.49	1.65	1.82	7
75	1041.0851	1.80	8.00	9.12	9
76	1251.2675	2.69	10.34	12.88	5
77	778.9151	1.33	5.93	8.17	28
78	540.0895	1.49	6.36	7.51	8
79	879.7062	0.38	1.73	1.97	4
80	660.4645	1.24	5.33	6.30	16
81	1018.4865	0.51	2.07	2.27	3
82	1559.9840	0.81	4.47	4.99	3
83	510.4223	0.29	1.14	1.34	3
84	1149.9670	0.71	3.16	3.68	6
85	1209.2972	0.30	1.20	1.44	3
86	347.7717	0.00	0.00	0.00	1
87	664.0903	3.31	9.20	11.62	5
88	957.6388	3.02	13.41	15.64	12
89	928.9954	2.80	10.11	12.53	22
90	560.1024	0.78	4.02	7.04	11
91	229.2288	1.42	5.90	7.01	6
92	606.1797	1.05	5.77	7.73	25
93	215.3416	2.30	8.38	11.33	37
94	273.1755	0.00	0.00	0.00	1
95	1218.1878	0.00	0.00	0.00	1
96	2082.3288	1.46	7.71	8.06	4
97	201.3224	1.57	6.16	8.20	19
99	886.7613	1.29	6.12	7.48	11
100	1045.0343	0.00	0.00	0.00	1
101	−109.6269	1.35	5.24	5.87	3
102	111.0590	2.43	8.89	12.58	8
103	1573.3769	0.56	2.89	2.90	2
104	1483.1289	0.43	2.03	2.03	2
105	1506.8136	1.23	6.50	7.89	3
106	484.6371	1.91	9.28	10.29	4
107	1379.4485	1.34	7.51	7.51	2
108	659.7336	1.57	6.51	7.25	3
109	492.0707	0.94	3.91	4.73	4
111	971.0365	0.00	0.00	0.00	1
113	428.8911	0.40	1.70	1.81	4
115	612.9506	1.17	5.07	5.15	3
116	562.1791	0.00	0.00	0.00	1
117	761.6006	1.29	5.49	5.53	2.00

$f(T_b)$ contained the number of atoms in the molecule and three parameters, which were regressed in a non-linear outer loop. The procedure was described in detail in the previous work.

While this method is very fast and does not require any initial values for the group contributions, it is not possible to regress the contributions of individual groups separately, as this would lead to different and incompatible values of

the constant a_0 . This was the reason that any regression was performed on the full set of data and why it was difficult to identify unreliable data, inappropriate group definitions, etc. In this work, we therefore used a linear regression³ of the

³ Details concerning the mathematical procedure can be found at www.property-estimation.org.

Table 4

Second-order contributions, number of components used for regressing these values and mean absolute deviation in temperature for these components

Group number	Group contribution (K)	Mean absolute error (%)	Mean absolute error (K)	Standard deviation (K)	Number of components
118	40.4205	1.20	6.04	8.22	135
119	−82.2328	2.96	12.38	13.88	19
120	−247.8893	0.16	0.43	0.43	2
121	−20.3996	2.35	8.33	10.70	139
122	15.4720	2.25	8.41	10.87	69
123	−172.4201	2.00	6.85	9.07	99
124	−99.8035	2.03	6.77	9.49	37
125	−62.3740	1.84	6.54	8.92	52
126	−40.0058	1.52	6.42	8.61	180
127	−27.2705	1.32	6.63	8.90	81
128	−3.5075	1.51	7.24	9.68	84
129	16.1061	1.38	6.82	8.84	101
130	25.8348	1.43	6.57	8.30	27
131	35.8330	1.20	5.37	7.08	88
132	51.9098	0.99	4.42	6.11	44
133	111.8372	0.92	4.29	6.58	18

type

$$y_i = \sum_{j=1}^M a_j x_{ij} \quad (2)$$

After implementing the new algorithm, model development started with regression of group contributions for saturated, unsaturated and aromatic hydrocarbons.

3.2. Regression of group contributions for saturated hydrocarbons

In the previous work, we used only seven group parameters for saturated hydrocarbons. A detailed analysis revealed rather high deviations for some, but not all highly branched isomers. One way to improve this would have been to introduce several larger groups or second-order corrections, but, as it was the goal to be as economical as possible with the number of group parameters, other options were tried. Introducing larger groups can greatly reduce errors for certain components in the available database but may lead to large errors when estimating properties of new components, which would require yet another large group correction.

The groups defined in the method contain only information on one carbon atom plus the information, whether it is part of a chain or a ring. To be able to describe isomer effects, a parameter must be introduced, which contains information about the greater neighborhood of a carbon. From the analysis of our regression results, it became apparent, that the number of carbons connected to the two neighbors of a C–C-bond would contain such information. An example with six bond neighbors is given in Fig. 1.

The strength of molecular interactions leading to ΔH_v of hydrocarbons was considered to be nearly solely due to the size of the molecular surface. Higher branched alkanes have a smaller molecular surface, so it was reasonable to expect lower boiling points for more steric hindered alkanes. In

contrary, significantly higher boiling points were found. In case of 2,2,3,3-tetramethylpentane the normal boiling temperature is 413.4 K compared to only 395.5 K for 2,2,4,4-tetramethylpentane.

In a MOPAC ab-initio calculation higher polarizabilities were found for the electrons in the higher boiling compound suggesting higher electron delocalization and stronger dispersive forces. Vapor pressure data from literature were regressed for both components and ΔH_v as well as ΔS_v were calculated. A difference of less than 0.8% in ΔS_v was found, whereby the larger part of this difference results from the higher translational entropy in the vapor phase due to the larger molar volume at the higher normal boiling temperature. On the other hand, the difference in ΔH_v was much more pronounced (5.4%), which supports the view of stronger dispersive forces in the liquid phase.

To improve the prediction of the normal boiling point of isomers, three steric corrections were introduced:

- (1) Both C of a C–C-bond have three further C neighbors (C_3C-CC_3).
- (2) One C of a C–C-bond has three and the other one has two further C neighbors (C_3C-CC_2).

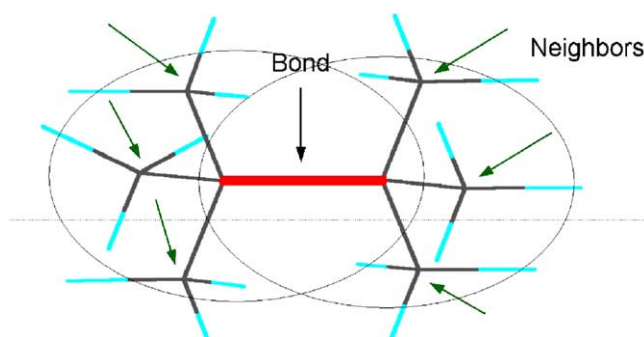


Fig. 1. Steric contribution from the number of carbons around a C–C-bond.

Table 5

Group interaction contributions, interacting groups (see Table 9) number of components used for regressing these values and mean absolute deviation in temperature for these components

Group number	Interacting groups	Group contribution (K)	Mean absolute error (%)	Mean absolute error (K)	Standard deviation (K)	Number of components
135	A–A	291.7985	1.89	9.76	12.57	37
136	A–M	314.6126	1.24	5.73	7.42	8
137	A–N	286.9698	1.20	6.06	6.77	6
138	A–L	38.6974	0.00	0.00	0.00	1
139	A–C	146.7286	0.96	4.63	4.64	2
140	A–D	135.3991	1.79	8.59	10.30	52
141	A–E	226.4980	0.00	0.00	0.00	1
142	A–F	211.6814	2.49	11.46	13.19	18
143	A–G	46.3754	1.67	7.82	8.66	8
144	A–J	–74.0193	0.38	1.99	2.26	4
145	A–P	306.3979	1.31	5.87	6.36	3
146	A–I	435.0923	0.00	0.00	0.00	1
147	A–S	1334.6747	0.00	0.00	0.00	1
148	B–B	288.6155	1.04	5.50	6.01	4
149	B–M	797.4327	0.00	0.00	0.00	1
150	B–C	–1477.9671	0.00	0.00	0.00	1
151	B–D	130.3742	2.60	13.44	15.25	10
152	B–F	–1184.9784	0.00	0.00	0.00	1
153	B–G	–	–	–	–	–
154	B–H	43.9722	0.00	0.00	0.00	1
155	B–Q	–1048.1236	1.07	5.34	5.69	3
156	B–S	–614.3624	2.44	13.09	14.36	3
157	M–M	174.0258	0.95	4.21	5.10	15
158	M–N	510.3473	0.98	5.04	5.44	4
159	M–D	124.3549	1.18	5.87	7.15	10
160	M–F	182.6291	1.79	9.89	10.54	3
161	M–J	–562.3061	0.78	4.00	4.00	2
162	M–Q	663.8009	1.60	9.10	9.10	2
163	M–I	395.4093	0.00	0.00	0.00	1
164	M–S	27.2735	1.71	8.36	8.36	2
165	N–N	239.8076	0.55	2.69	3.76	6
166	N–D	101.8475	0.56	2.34	2.64	6
167	N–F	317.0200	0.00	0.00	0.00	1
168	N–G	–215.3532	0.00	0.00	0.00	1
169	N–S	758.9855	0.00	0.00	0.00	1
170	L–L	217.6360	0.88	4.35	5.66	10
171	L–AF	501.2778	0.00	0.00	0.00	1
172	C–C	117.2044	1.46	8.38	10.28	4
173	C–D	612.8821	1.55	8.67	12.57	6
174	C–F	–183.2986	2.17	12.12	12.13	2
175	C–G	–55.9871	4.62	22.27	22.34	2
176	O–O	–356.5017	0.75	4.34	4.76	3
177	O–Q	–263.0807				3
178	D–D	91.4997	1.70	7.72	10.43	185
179	D–E	178.7845	1.94	9.06	9.86	3
180	D–F	322.5671	1.89	9.17	12.53	26
181	D–G	15.6980	1.44	7.16	8.28	10
182	D–H	17.0400	2.50	10.79	12.78	12
183	D–J	394.5505	1.57	7.68	7.77	4
184	D–Q	963.6518	0.68	3.74	3.74	2
185	D–P	293.5974	0.56	2.66	3.00	3
186	D–I	329.0050	0.33	1.29	1.29	2
187	E–E	1006.3880	0.00	0.00	0.00	1
188	E–H	163.5475	0.00	0.00	0.00	1
189	F–F	431.0990	2.04	10.25	12.78	70
190	F–G	22.5208	2.41	11.49	13.86	25
191	F–Q	–205.6165	0.00	0.00	0.00	1
192	F–P	517.0677	2.66	13.29	13.85	6

Table 5 (Continued)

Group number	Interacting groups	Group contribution (K)	Mean absolute error (%)	Mean absolute error (K)	Standard deviation (K)	Number of components
193	F–I	707.9404	0.96	4.39	5.29	4
194	G–G	–303.9653	2.47	12.03	14.06	10
195	G–H	–391.3690	0.00	0.00	0.00	1
196	G–Q	–3628.9026				
197	G–K	381.0107	0.45	2.29	2.45	3
198	G–P	–574.2230	0.00	0.00	0.00	1
199	G–I	176.5481	0.31	1.51	1.74	3
200	G–S	124.1943	0.85	4.28	4.67	3
201	H–H	582.1763	0.00	0.00	0.00	1
202	H–Q	140.9644	0.08	0.46	0.46	2
203	H–K	397.5750	0.00	0.00	0.00	1
204	H–I	674.6858	0.00	0.00	0.00	1
205	J–J	–11.9406	1.65	7.86	9.42	7
206	Q–Q	65.1432	0.91	5.43	5.95	3
207	K–P	–101.2319	0.00	0.00	0.00	1
208	K–R	–348.7400	2.09	9.21	10.14	10
209	P–S	–370.9729	0.00	0.00	0.00	1
210	I–R	–888.6123	0.00	0.00	0.00	1
211	R–R	–	–	–	–	–
212	S–S	–271.9449	1.84	7.47	8.25	3

(3) Both C of a C–C-bond have two further C neighbors (C₂C–CC₂).

In the case of C₃C–CC₁, no similar steric effect was found. Table 6 shows the improvements obtained for compounds, where these steric corrections were applied. For these highly branched compounds, the other methods give rather large deviations, even when using a larger number of parameters or corrections (CG, MP).

Altogether seven groups and corrections are used for non-cyclic saturated hydrocarbons. In comparison JR, SB, CG, MP, CR and EWOR used 4, 7, 9, 10, 4, and 5, respectively.

Table 7 shows the results obtained for hydrocarbons together with the results of previous methods. While the aver-

age deviations were only moderately reduced as compared to the previous work, significant improvements are mainly found for special cases as shown in Table 6.

3.3. Alkenes and alkynes, aromatics

When analyzing unsaturated non-aromatic hydrocarbons, an error in the group definition for the previous method was detected. Due to this error, cumulated dienes were fragmented as C=C–C. In the new method, the additional groups C=C=C, C=C–C=C (chain), C=C–C=C (ring) and C≡C–C≡C had to be introduced. In addition, a steric correction for highly branched alkenes was introduced

Table 6

Comparison of average deviations in case of steric corrections (AAD/K)

	Number of components	With correction	Without correction	JR	SB	CG	MP	CR	EW
C ₃ C–CC ₃	17	3.07	9.76	21.91	19.77	7.00	17.51	10.27	11.27
C ₃ C–CC ₂	28	5.38	7.61	20.80	17.89	11.23	12.09	9.68	15.60
C ₂ C–CC ₂	94	7.53	8.29	15.68	12.74	10.30	7.31	9.22	13.63

Table 7

Deviations of the models for different types of hydrocarbons

Compounds	AAD M (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Hydrocarbons (HC)	5.80	689	680	679	686	555	688	683	18.88	10.68	8.85	10.79	6.89	8.08
Saturated HC	4.92	266	266	266	266	266	266	266	20.05	14.18	7.87	9.22	6.63	8.90
n-Alkanes	5.54	27	27	27	27	27	27	27	55.69	12.10	18.67	13.64	6.47	12.83
Alkanes (non-cyclic)	4.95	192	192	192	192	192	192	192	25.14	16.36	8.49	7.18	6.68	8.76
Alkanes (cyclic)	4.85	74	74	74	74	74	74	74	6.85	8.53	6.26	14.52	6.51	9.28
Aromatic HC	6.02	177	177	167	177	115	177	176	29.12	7.27	12.04	9.03	6.70	7.98
Alkenes (HC)	6.55	180	173	180	180	126	180	180	9.18	8.53	6.16	17.54	7.42	7.90
Alkenes (cyclic C=C)(HC)	7.78	53	49	53	53	26	53	53	6.92	6.42	6.16	15.13	8.13	11.61
Alkynes (HC)	3.01	35	35	35	35	33	35	35	13.13	12.16	13.84	3.98	5.40	3.59

Table 8

Comparison of average deviations in case of the unsaturated steric correction ((C=)(C)C–CC₃) (AAD/K)

	Number of components	With correction	Without correction	JR	SB	CG	MP	CR	EW
((C=)(C)C–CC ₃)	27	6.57	7.35	33.39	11.51	13.08	12.09	8.86	13.42

((C=)(C)C–CC₃). The deviation for components, where this correction was applied, is given in Table 8.

No group corrections were required for aromatic compounds. The deviations for several groups of unsaturated and aromatic hydrocarbon compounds are given in Table 7.

3.4. Monofunctional compounds

As described in the previous work, high deviations were observed, for e.g. diols and triols. In this work, the regression of contributions for polar and hydrogen-bonding groups was for many cases first limited to components with one functional group only. In the case of simple substituents like halogens, where we observed a sufficient additivity of the group effect on the boiling point, the regression was not limited to components with a group frequency of one. Table 9 gives a list of all groups considered to be non-additive.

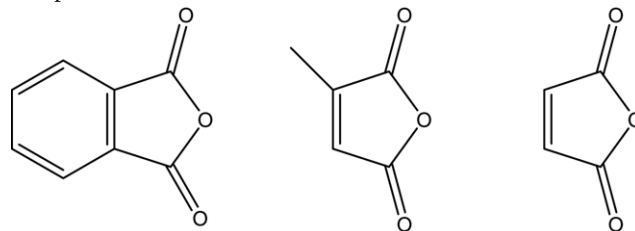
During the parameter regression, it became apparent that further differentiations in the structural group definitions had to be introduced. A special OH-group connected to a chain of less than five carbon atoms in the previous method is now applied to molecules containing less than five carbon atoms. The fluorine group used for fluorine connected to a carbon carrying already one fluorine or chlorine had to be replaced by two groups (one for fluorine as the second halogen and one for chlorine). Two different aldehyde and ketone groups are now used depending on whether the neighboring carbon is part of an aromatic system or not.

Table 9

Groups considered to be non-additive (group-ID(s) given in brackets (see also Table 1))

A	Alcohol (–OH) (34–37)
B	Phenol (–OH(a)) (37)
C	Carboxylic acid (–COOH) (44)
D	Ether (–O–) (38)
E	Epoxide (> (OC2)<) (39)
F	Ester (–COOC–) (45, 46, 47)
G	Ketone (–CO–) (51, 92)
H	Aldehyde (–CHO) (52, 90)
I	Aromatic oxygen (–O(a)–) (65)
J	Sulfide (–S(na)–) (54)
K	Aromatic sulfur (–S(a)–) (56)
L	Thiol (–SH) (53)
M	Primary amine (–NH ₂) (40, 41)
N	Secondary amine (> NH) (42, 97)
O	Isocyanate (–OCN) (80)
P	Cyanide (–CN) (57)
Q	Nitrate (69)
R	Aromatic N in five-ring (=N(a)–(r5)) (66)
S	Aromatic N in six-ring (=N(a)–(r6)) (67)

In case of carboxylic acid anhydrides, a new group for cyclic anhydrides containing an aromatic or double bond in the ring (connected to sp₂ carbons) was added to describe components like



Instead of one secondary amine group, there are now two groups for aromatic and non-aromatic secondary amines. The group for silicon with an electronegative neighbor was split into two separate groups, one with oxygen as a neighbor, and one for fluorine or chlorine.

A number of groups were added to extend the applicability of the method. All new and modified groups are compiled in Table 10.

In the case of aldehydes, ketones, carboxylic acids and esters, larger deviations were observed for molecules with a carbon–carbon π -bond in conjugation with the carbonyl double bond. As the effect was similar for all four classes of compounds, a common correction for C=C–C=O was introduced. The correction takes into account the ability of the electronegative oxygen to polarize the electrons in the conjugated system resulting in a significantly larger charge separation than in case of the isolated carbonyl double bond. It is not of importance, whether the C=C-bond is part of an aromatic system or not.

No further special problems were observed with monofunctional compounds. Tables 11–15 show the deviations for the different types of components. This extensive comparison provides valuable information about the model as it is possible to judge not only, which estimation method gives the best results but also, which typical deviations have to be expected. For this reason, the program package ARTIST used for the development of this method was extended to automatically detect the types of compounds a molecule belongs to and report the deviations of these types for the different methods in order to assist the user in selecting the appropriate method for the estimation. Similar tables were generated for other properties (e.g. critical data).

3.5. Multifunctional compounds

As was already discussed in the previous publication for the case of diols, all group contribution methods more or less fail to predict properties for components with more than one strongly associating (see Table 9) functional group.

Table 10
Group changes with respect to the previous method

Periodic group	Structure	Previous name	New name	Comments
17	–F	F–(C–([F, Cl]))–b	F–(C–([F]))–b	Fluorine connected to a carbon, which already carries one fluorine
	–F	New	F–(C–([Cl]))–b	Fluorine connected to a carbon, which already carries one chlorine
16	–OH	–OH short chain	–OH short chain< C ₅	Now only applies to molecules with less than five carbon atoms
	–CHO	CHO–(C)	CHO–(Cna)	Now only applies when connected to a non-aromatic carbon
	–CHO	New	CHO–(Ca)	Aldehyde group connected to an aromatic carbon
	> C=O	O=C< (C)2	O=C< (Cna)2	Ketone group connected to a non-aromatic carbon
	> C=O	New	O=C< (Ca)2	Ketone group connected to an aromatic carbon
	–CO–O–CO–	–CO–O–CO–	–CO–O–CO–	Cyclic anhydride connected to sp ₃ carbons
	–CO–O–CO–	New	–CO–O–CO–(sp ₂)	Cyclic anhydride connected to sp ₂ carbons
	–OCOO–	New	–OCOO–	Carbonate
	> N(C=O)–	New	> N(C=O)–	Carbonyl with nitrogen as neighbor
	–O–O–	New	–O–O–	Peroxide
	> SO ₄	New	> SO ₄	Sulfate
	–SO ₂ N<	New	–SO ₂ N<	Sulfon amide
	> S=O	New	> S=O	Sulfoxide
	–O–CO–N<	New	–O–CO–N<	Carbamate
	> N–CO–N<	New	> N–CO–N<	Urea
	> Se<	New	> Se<	Selenium
13	> Al<	New	> Al<	Aluminum
15	–NH–	(C, Si)–NH–(C, Si)	(C, Si(na))–NH–(C, Si(na))	Secondary non-aromatic amine
	–NH–	New	(C, Si)–N(a)H–(C, Si)	Secondary aromatic amine
	NH=	New	NH=(C, Si)	Double bonded non-aromatic amine
	> N<	New	(C)> N< (C)	Quaternary amine
	–C≡N	New	(N)–C≡N	Cyanide connected to nitrogen
	ON=	New	–ON=	Isoazole
	SC≡N	New	(S)–C≡N	Thiocyanate
	CNCNC-ring	New	...=CNC=NC=...	Imadizole
	> P<	New	> P<	Phosphine
	–PO ₃	New	P(O–, O–, O–)	Phosphite
14	> C=C=C<	New	> C=C=C<	Cumulated double bond
	> C=C–C=C<	New	(r)> C=C–C=C< (r)	Conjugated double bond in a ring
	> C=C–C=C<	New	(c)> C=C–C=C< (c)	Conjugated double bond in a chain
	–C≡C–C≡C–	New	–C≡C–C≡C–	Conjugated triple bond
	> Si<	> Si< (e)	> Si< (O)	Silicon attached to oxygen
	> Si<	New	> Si< (F, Cl)	Silicon attached to F or Cl

The reason for this is, that for these groups no simple additivity is observed. In this work, the group contribution parameters were therefore regressed using only monofunctional compounds and additional group interaction parameters were introduced for multi-functional compounds. In contrary to simple monofunctional compounds, the size of the group interaction decreases with the size of the molecule. To take this into account, the sum of group interactions is divided by the number of non-hydrogen atoms in the molecule. A calculation example for a multi-functional compound is given in Table 17b and Table 16 shows the deviations of the different models for several types of multifunctional compounds.

The total group interaction GI can be calculated from the equation

$$GI = \frac{1}{n} \sum_{i=1}^m \sum_{j=1}^m \frac{C_{i-j}}{m-1} \quad (\text{where } C_{i-j} = C_{j-i}) \quad (3)$$

where GI is the total group interaction contribution, C_{i-j} is the group interaction contribution between group i and group j [K] ($C_{i-i} = 0$), n is the number of atoms in the molecule (except hydrogen), and m is the total number of interaction groups.

In case of a glycerol monoester (two OH-groups, one ester-group), this results in $\text{COH-OH} + 2\text{COH-ester}$.

3.6. Intramolecular hydrogen-bonding

While the method proposed here yields very good results for nearly all kinds of compounds, a few cases require some caution.

In cases, where steric effects force an intramolecular hydrogen-bond, this bond is not available anymore for interaction with another molecule and the boiling temperature is lower than expected. Typical cases are (a) *o*-nitrophenols, (b) *o*-hydroxy carbonyl aromats (e.g. salicylic acid), (c) *o*-alkoxy

Table 11

Comparison of the method proposed for monofunctional compounds (oxygen)

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Alcohols	6.14	150	150	148	150	131	150	148	25.57	6.57	12.35	13.24	6.77	13.73
Primary alkanols	6.52	46	46	46	46	43	46	46	20.29	6.78	9.61	14.16	7.29	15.77
Secondary alkanols	5.94	45	45	44	45	44	45	43	25.54	5.32	11.54	13.19	5.85	12.93
Tertiary alkanols	4.56	31	31	31	31	18	31	31	39.97	6.76	19.88	17.90	6.65	15.11
Diols, triols . . .	8.43	22	22	22	22	22	22	22	32.05	26.78	28.12	18.79	16.79	20.70
Ethers (HC)	4.94	98	97	98	78	77	98	93	9.90	6.61	8.93	5.22	5.78	8.37
Epoxides (HC)	3.29	12	12	12	10	9	12	8	12.07	9.61	28.29	5.62	7.12	7.63
Aldehydes (HC, no = – =O)	4.64	16	16	16	16	15	16	16	8.44	6.31	8.08	2.74	9.83	4.70
Aldehydes (HC, = – =O)	6.94	10	10	10	10	5	10	10	22.72	10.02	11.83	22.54	6.13	9.46
Ketones(HC, no = – =O)	4.92	50	49	50	46	41	50	50	14.05	7.49	8.30	5.68	6.53	10.03
Ketones(HC, = – =O)	8.28	16	11	15	11		16	13	11.71	12.07	7.12		9.95	14.47
Non-cyclic carbonates	2.12	6	6	6	4	4	4	4	48.86	53.94	3.46	5.50	3.65	73.82
Carboxylic acids (HC, no = – =O)	6.53	27	27	27	27	26	27	27	41.17	6.47	12.96	11.07	6.40	25.29
Carboxylic acids (HC, = – =O)	6.48	7	7	7	7	5	7	7	9.75	9.30	15.27	18.89	7.81	11.16
Esters (HC, no = – =O)	4.35	89	89	89	89	71	89	89	18.00	3.95	7.03	6.69	5.23	24.08
Esters (HC, = – =O)	4.09	21	21	21	14	16	21	21	10.64	5.64	11.78	7.47	5.46	7.29
Formic acids esters (HC)	2.34	17		17	17	13	17	17		64.48	8.71	2.43	4.82	4.05
Lactones (HC)	8.36	3	3	3	2		3	2	107.3	20.07	104.0		4.32	110.04

Table 12

Comparison of the method proposed for monofunctional compounds (halogen)

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Fluorinated saturated (HC)	6.37	64	64	64	64	63	64	1	18.53	59.89	25.62	14.25	7.35	5.54
Fluorinated (HC)	7.16	86	86	86	86	83	86	14	18.01	48.96	26.46	12.83	8.02	6.53
Chlorinated saturated (HC)	7.53	64	64	64	60		64	49	26.01	14.72	8.26		7.43	9.72
Chlorinated (HC)	6.96	117	117	117	95		117	98	21.36	14.05	9.36		6.62	7.94
Brominated saturated (HC)	5.48	31	31	31	31		31	28	14.76	8.82	6.91		6.34	16.93
Brominated (HC)	5.81	49	49	49	49		49	44	12.89	8.88	7.48		7.36	17.98
Iodinated (HC)	5.14	18	18	18	18	15	18	17	13.63	6.91	5.77	12.06	5.20	13.43
Halogenated compounds	8.09	565	514	520	430	132	564	382	22.63	23.15	18.86	12.83	9.09	20.69

Table 13

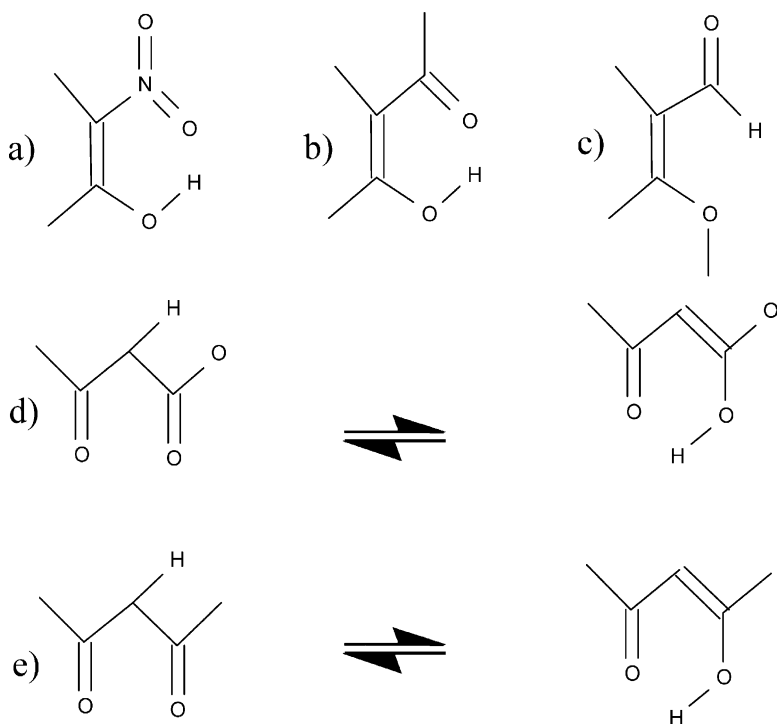
Comparison of the method proposed for monofunctional compounds (sulfur)

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Disulfides (HC)	2.96	4	4		4	3	4	3	7.96		20.17	1.12	3.38	2.73
Thiols (HC)	3.64	37	37	37	18	18	37	20	12.91	7.21	7.60	7.46	5.13	9.05
Thioether (HC)	4.85	30	30	30	28	19	30	27	15.45	8.05	11.49	3.88	4.56	7.13
Aromatic thioether (HC)	3.68	10	10		8	3	10	9	11.04		8.68	0.25	5.17	10.71
Sulfolane O=S=O	4.47	3					3						12.11	
Thiocyanates and isothio- cyanates	2.07	3					3						8.61	
Sulfates, sulfon amides and sulfoxide	5.22	6												
Sulfur compounds	4.08	93	81	67	58	43	87	59	13.38	7.59	10.49	4.93	5.22	8.10

Table 14
Comparison of the method proposed for monofunctional compounds (nitrogen)

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Primary amines	5.06	43	43	42	41	37	43	43	20.23	7.22	13.14	4.66	7.94	8.40
Secondary amines	4.75	31	31	31	31	25	31	31	14.18	9.17	13.14	5.24	7.32	17.25
Tertiary amines	7.10	13	13	13	13	11	13	13	13.62	9.45	8.82	5.48	7.57	8.63
Five-membered N rings	5.39	12	9	12		1	12	3	44.52	29.27		0.11	7.14	56.15
Six-membered N rings	7.49	28	28	28	20	14	28	24	13.20	7.52	21.60	4.13	8.99	12.64
Cyanides and cyanamides	4.83	26	26	26	19	23	26	23	22.42	3.38	7.91	8.45	10.09	24.76
Amides	7.40	4	3	4	3		4	3	111.0	47.77	8.04		8.58	15.23
Mono amides	7.53	6	4	6			6	5	83.59	22.77			12.51	12.36
Diamides	5.64	8	2	8	5		8	7	74.45	11.38	19.16		6.80	6.64
Carbonyl with N attached	3.91	4	4	4	2				27.64	64.83	42.72			
Isocyanates	5.41	9		9			9			69.94			9.90	
Oximes	8.00	9					9						7.88	
All nitrites (HC)	5.24	15	8	8	7	1	15	8	28.81	12.37	6.03	0.01	7.82	7.18
Nitrates (HC)	2.96	5					5						3.83	

benzaldehydes, (d) *b*-keto esters and (e) 1,3-dicarbonyl compounds:



No special correction for intramolecular hydrogen-bonding was introduced into the model and the few cases, where this effect was observed were removed from the database for regression and model testing.

3.7. Database

Normal boiling temperatures for approximately 2850 components from the Dortmund Data Bank (DDB) were used, all of which are constructed of the structural groups

proposed in this work. The greater part of this data was entered after work on the data bank started in 1973 and were extensively used, e.g. the calculation of phase equilibrium data. This is the reason why this set of data may be regarded as very reliable. Additional data were added from the Beilstein database for specific structures of interest. In addition, molecular structures for approximately 16,000 components were stored in form of connection tables, so that the molecules could easily be fragmented into groups by the automatic procedure described before [11]. The frequency of occurrence of

Table 15
Comparison of the method proposed for monofunctional compounds (various)

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
Phosphates	5.05	4					4						6.98	
Arsine	3.16	6					6						3.07	
Germanium	0.00	1					1						15.81	
Germanium + Cl3	1.20	3					3						6.03	
Stannium	1.14	3		3			3			2.42			1.30	
Boron	5.80	7					7						4.25	
Silicon	4.44	27		23			27	3		22.18			4.34	0.62
Silicon to oxygen	10.76	43					40	17					10.80	17.11
Silicon to F and Cl	8.38	37					37	4					9.37	0.78
Anhydrides chains	9.11	4	4	4	3		4	2	19.21	39.44	10.23		15.62	5.55
Anhydrides cyclic, aromatic	7.30	3	3	3	1		3		37.15	53.92	69.52		38.55	
Acid chloride	3.33	16	12	16			16	1	10.95	55.34			3.97	144.82
Aromatic oxygen	4.60	5	5			2	5	2	22.18			8.61	7.36	23.36
Carbamate	4.89	6	4	6	5				20.19	48.58	28.06			
Urea (HC)	0.00	1	1	1	1				56.90	5.44	35.58			
Imidazol (Ar-five-ring)	9.28	4		4						62.53				
Nitrogen bonded to four carbons	5.24	3												
Selenium	0.00	1		1						10.76				
Aluminum	5.49	2												

the individual structural groups and second-order corrections in a database of 2850 components is given in Tables 1 and 3, respectively.

3.8. Test of the predictive capability

In order to test the predictive capability of the method, experimental normal boiling temperatures for 199 components, not in the database used for regression, were compared with the predicted values. This led to a mean absolute deviation in temperature of 6.37 K as compared to 7.83 K for the previous method (JR: 20.31 K, SB: 8.28 K, CG: 13.02 K, MP:

12.03 K, EWOR: 14.40 K). Components were only selected to which all methods could be applied. As components with group interactions were urgently needed for the model development, these components are not present in the test set. The deviation obtained for this latter test set is nearly identical to the deviation in case of the prior test set (6.52 K).

4. Results and discussion

The method proposed here gives an absolute average deviation of 6.52 K for 2812 compounds (JR: 21.42 K (2506), SB:

Table 16
Comparison of the method proposed for multifunctional compounds

Compounds	AAD (this work)	Components	Number of components						Absolute average deviations (K)					
			JR	SB	CG	MP	CR	EW	JR	SB	CG	MP	CR	EW
OH—OH	10.52	27	26	26	27	25	27	27	30.26	24.97	26.12	18.18	15.91	20.45
OH—NH ₂	4.71	6	6	6	5	5	6	5	23.55	23.27	22.66	15.24	15.21	26.23
OH—etherO	7.56	32	31	31	28	26	31	31	23.65	6.52	16.30	12.04	7.71	10.26
OH—ester	11.38	17	16	17	17	11	17	17	32.71	12.59	13.51	14.01	12.46	40.56
OH—ketone	7.82	8	7	8	7	7	8	6	17.03	10.82	12.44	9.51	6.81	46.41
OH(a)—etherO	14.93	9	9	9	9	8	9	9	20.91	14.62	16.16	16.70	13.64	18.51
NH ₂ —NH ₂	3.98	13	13	13	12	11	13	12	17.73	9.53	7.70	7.54	5.18	5.52
NH ₂ —etherO	4.40	7	7	7	7	7	7	7	8.86	9.22	15.06	9.62	6.28	9.26
SH—SH	4.35	10	10	10	10	10	10	10	32.66	7.91	32.15	22.61	5.67	8.52
COOH—etherO	8.67	6	6	6	6	6	6	6	18.52	16.17	36.20	14.30	17.07	92.44
EtherO—etherO	7.45	160	129	132	86	95	160	108	14.97	11.98	17.22	8.86	8.15	14.59
EtherO—ester	8.37	23	22	23	21	19	23	23	19.62	13.05	9.05	9.57	9.73	24.63
EtherO—ketone	7.22	8	4	8	8	4	8	7	13.73	10.68	10.06	3.82	8.35	55.98
EtherO—alde	11.61	10	10	10	9	7	10	9	23.66	15.69	9.42	9.01	10.90	41.11
Ester—ester	9.51	66	66	66	66	49	66	66	44.23	12.24	15.22	14.62	11.58	59.00
Ester—ketone	10.46	23	20	23	18	14	23	18	20.98	12.24	11.32	8.14	10.25	45.31
Ester—CN	13.29	6	6	6	2	4	6	1	35.28	19.58	10.72	16.40	14.74	8.76
Ketone—ketone	12.03	10	9	10	8	7	10	8	28.58	15.76	12.21	15.52	16.87	86.13
Teth—teth	7.86	7	7	7	4	2	5		28.19	29.14	17.16	7.82	9.42	

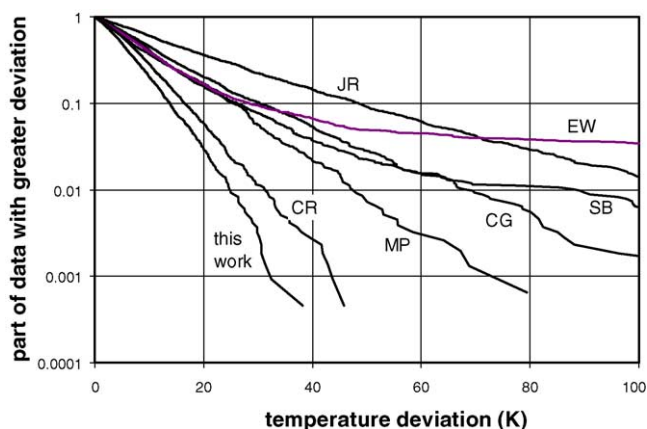
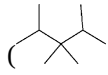


Fig. 2. Fraction of the data with deviations larger than a given temperature based on 2171 data points (MP: 1539, EW: 2035).

14.46 K (2579), GC: 13.25 K (2259), MP: 10.23 K (1665), CR: 8.16 (2758), EWOR: 16.82 (2280)). This means that it combines the lowest deviation with the broadest range of applicability.

The probability of a prediction failure (extreme deviation between experimental and estimated value) was chosen as the most important criterion for the reliability of a model. Fig. 2 shows a plot of the fraction of data with a deviation larger than a certain temperature versus the deviation in temperature. Calculations are based on a common set of 2169 components, for which all models except MP and EWOR can estimate the normal boiling point. The curve for the latter methods represents results for only 1537 resp. 2033 compounds. It can be seen, that in case of the new method, only 3% (previous work 6%) of the data are estimated with a deviation larger than 20 K. The other models range between 16% (Stein, Braun) and 36% (Constantinou, Gani).

Large deviations are observed in the case of molecules, which consist solely, or mainly of one structural group like 1,3-butadiene, benzene or cyclopropane (23.9 K), cyclobutane (18.4 K), cyclopentane (12.7 K) and cyclohexane (19.2 K) or which represent the first elements in a homologous series like ethanol (−18.4 K). This was to be expected, as it is commonly known, that the first member(s) of a homologous series in most cases disobey the general trend. Luckily, experimental normal boiling points for these simple substances are generally available and there is no need for a group contribution estimation.

In the case of 2,3,3,4-tetramethylpentane () there are two steric corrections with a common carbon. Due to the limited database, no special correction was introduced for this case (−26.4 K, too low).

Other causes of large deviations include intramolecular hydrogen-bonding (discussed above) or strong proximity effects like in trichloroacetic acid (−16.6 K).

5. Examples

To illustrate the application of the proposed method, a detailed procedure for the estimation of T_b is given in Tables 17a–d for

- 3,3,4,4-tetramethylhexane illustrating the application of the steric correction;
- di-isopropanolamine illustrating the application of group interactions;
- perfluoro-2-propanone illustrating the application of halogen corrections; and
- methyl *m*-toluate illustrating the application of the $C=C-C=O$ correction.

Table 17a
Estimation of the normal boiling temperature of 3,3,4,4-tetramethylhexane

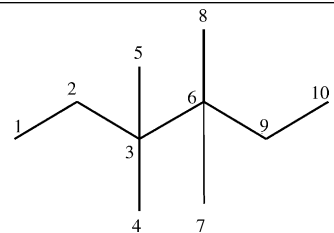
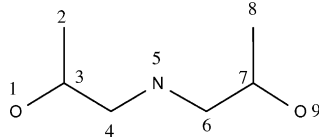
				
Component: 3,3,4,4-tetramethylhexane, number of atoms: 10				
Group	Atoms	Frequency	Contribution	Total
1	1, 4, 5, 7, 8, 10	6	177.307	1063.840
4	2, 9	2	239.453	478.906
6	3, 6	2	249.581	499.162
Steric corrections				
Group	Bond	Frequency	Contribution	Total
C_3C-CC_3 (133)	3–6	1	111.837	111.837
Sum				2153.745
$T_b = \frac{2153.745K}{10^{0.6583 + 1.6868}} + 84.3395K = 429.5K$. Experimental $T_b = 444.0 K$.				

Table 17b

Estimation of the normal boiling temperature of di-isopropanolamine



Component: di-isopropanolamine, number of atoms: 9

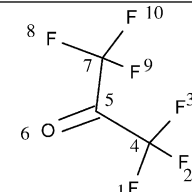
Group	Atoms	Frequency	Contribution	Total
1	2, 3	2	177.307	354.613
7	3, 4, 6, 7	4	266.877	1067.507
34	1, 9	2	390.245	780.489
42	5	1	233.099	233.099
Interactions				
Group	Atoms	Frequency	Contribution	Total
OH–OH (135)	1–9	(1 ^a)/9	291.799	32.422
OH–NH (137)	1–5, 9–5	(2 ^a)/9	286.970	63.771
Sum				2521.902

$$T_b = \frac{2521.902K}{90.6583 + 1.6868} + 84.3395K = 509.3K. \text{ Experimental } T_b = 522.0K. m = 3(OH_1 + OH_2 + NH).$$

$$^a \text{ From Eq. (3): } GI = (C_{OH_1-OH_9} + C_{OH_1-NH_5} + C_{NH_5-OH_1} + C_{NH_5-OH_9} + C_{OH_9-OH_1} + C_{OH_9-NH_5})/n(m-1) = (2C_{OH-OH} + 4C_{OH_1-NH_5})/n(m-1) = (C_{OH-OH} + 2C_{OH_1-NH_5})/n$$

Table 17c

Estimation of the normal boiling temperature of perfluoro-2-propanone

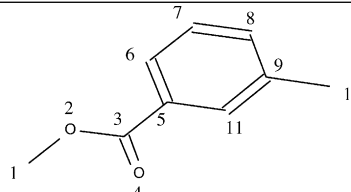


Component: perfluoro-2-propanone, number of atoms: 10

Group	Atoms	Frequency	Contribution	Total
7	4, 7	2	266.877	533.754
21	1, 2, 3, 8, 9, 10	6	53.187	319.123
51	5, 6	1	618.978	618.978
Corrections				
Group	Atoms	Frequency	Contribution	Total
120	5	1	–247.889	–247.889
121	4, 7	2	–20.400	–40.800
123	–	1	–172.420	–172.420
Sum				1010.746

$$T_b = \frac{1010.746K}{100.6583 + 1.6868} + 84.3395K = 246.3K. \text{ Experimental } T_b = 245.9K.$$

Table 17d

Estimation of the normal boiling temperature of methyl *m*-toluate


Component: methyl *m*-toluate, number of atoms: 11

Group	Atoms	Frequency	Contribution	Total
2	1	1	251.834	251.834
3	10	1	157.953	157.953
15	6, 7, 8, 11	4	235.346	941.385
16	5, 9	2	315.413	630.826
45	2, 3, 4	1	636.202	636.202
Corrections				
Group	Atoms	Frequency	Contribution	Total
128	5 and 9	1	–3.508	–3.508
118	3–5–6	1	40.421	40.421
Sum				2655.112

$$T_b = \frac{2655.112K}{110.6583 + 1.6868} + 84.3395K = 490.6K. \text{ Experimental } T_b = 492.3K.$$

6. Conclusion

A group contribution method for the estimation of the normal boiling point of non-electrolyte organic compounds was developed. The predictions are based exclusively on the molecular structure of the compound. The results of the new method are in most cases far more accurate and never significantly worse than previous methods. Structural groups were defined in a standardized form and the fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions.

Second-order corrections were limited to those cases, in which larger structures or structural effects could not be defined as structural groups (e.g. *o*-, *m*- and *p*-positions at rings, three-, four- and five-membered rings and molecules with only one or two hydrogen atoms).

Structural groups were usually defined including the neighboring atoms thus implementing knowledge about the electronic structure in the respective context.

Currently the database for normal boiling points is being extended. In cases, where only low-pressure data are available, these are used for extrapolation. Once a significant amount of new data is available, the method will be updated and further extended.

To assist the user in estimating the probable error for a specific boiling point estimation, detailed tables with deviations for a large number of classes of chemically similar compounds were stored in computer readable form and the classes a new compound belongs to are automatically derived from the structure.

In addition, work has started to use the same approach for the prediction of other pure component properties.

Most recent updates and corrections can be found at <http://www.property-estimation.org>.

List of symbols

a, b, c	adjustable parameters
c_{ij}	group interaction contribution between group i and group j [K] ($c_{ii} = 0$)
C	group contribution
GI	total group interaction contribution

H	enthalpy
N	number of groups
N	number of atoms except hydrogen
S	entropy
T	absolute Temperature
V	volume

Subscripts

b	boiling
c	critical
vap	vaporization

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References

- [1] W. Cordes, J. Rarey, Fluid Phase Equilib. 201 (2002) 409–433.
- [2] G.M. Barrow, Physical Chemistry, fourth ed., McGraw Hill, Singapore, 1985.
- [3] B.E. Poling, J.M. Prausnitz, J.P. O'Connell, The Properties of Gases and Liquids, fifth ed., McGraw Hill, New York, 2000.
- [4] R.S. Boethling, D. Mackay, Handbook of Property Estimation Methods for Chemicals, CRC Press, Boca Raton, FL, 2000.
- [5] L.R. Constantinou, R. Gani, AIChE J. 40 (10) (1994) 1697–1710.
- [6] K.G. Joback, R.C. Reid, Chem. Eng. Commun. 57 (1987) 233–243.
- [7] S.E. Stein, R.L. Brown, J. Chem. Inf. Comput. Sci. 34 (1994) 581–587.
- [8] J. Marrero-Morejon, E. Pardillo-Fontdevila, AIChE J. 45 (3) (1999) 615–621.
- [9] J. Marrero, R. Gani, Fluid Phase Equilib. 183–184 (2001) 183–208.
- [10] D. Ericksen, W.V. Wilding, J.L. Oscarson, R.L. Rowley, J. Chem. Eng. Data 47 (2002) 1293–1302.
- [11] W. Cordes, J. Rarey, F. Deliquie, J. Gmehling in: D. Ziessow (Ed.), Software Development in Chemistry 7, Proceedings of Computer in der Chemie 7, Springer-Verlag, Berlin, 1993.